

Multi-Contact 4C (MC-4C): long molecule sequencing of complex proximity ligation products to uncover local cooperative and competitive chromatin topologies

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Abstract

We present the protocol and data analysis toolbox for Multi-Contact 4C (MC-4C), a new proximity ligation method tailored to study the higher-order chromatin contact patterns of selected genomic sites. Conventional chromosome conformation capture (3C) methods fragment proximity ligation products for efficient analysis of pairwise DNA contacts. In contrast, MC-4C is designed to preserve and collect large concatemers of proximity ligated fragments for long molecule sequencing on Oxford Nanopore or Pacific Biosciences platforms, thus allowing study of multi-way chromatin interactions. By inverse PCR with primers specific for a fragment of interest (the viewpoint) and DNA size selection, sequencing is selectively targeted to thousands of different complex interactions containing this viewpoint. A tailored statistical analysis toolbox employing data-intrinsic background models then discerns whether contacts between more than two regulatory sequences are mutually exclusive or, conversely, simultaneously happening at chromatin hubs. The entire procedure can be completed in two weeks and requires access to a third generation sequencing platform.

Introduction

Development of MC-4C

Chromatin conformation capture (3C) methods ^{1,2}, employ in situ DNA digestion and proximity ligation to detect contacting DNA fragments and investigate chromosome topologies inside cells. Most protocols also include a formaldehyde crosslinking step prior to digestion to preserve spatial proximity, although fixation seems not strictly necessary ³. Current high-throughput 3C variant methods include Hi-C (for unbiased and untargated genome-wide contact analysis) ⁴, 4C and capture-C technologies (for unbiased analysis of contacts made by one or more selected genomic sites of interest) ⁵⁻⁷ and ChIA-PET and Hi-ChIP (for contact analysis between sites occupied by a protein of interest) ^{8,9}. All these methods use Illumina short read sequencing for massively parallel analysis of pairwise DNA contacts, which for optimal results requires trimming the ligation products to sizes far below one thousand base pairs. A limitation of all methods measuring pairwise contacts is that when three or more elements are identified to interact in a cell population, it remains unclear whether this occurs simultaneously in the same cell or in a pairwise fashion in different cells or at different time points. Pairwise methods thus also fail to identify if interactions occur (i) in a cooperative fashion (forming preferred multi-way contacts, or hubs, on the same allele), (ii) in an uncoordinated (random) fashion, or (iii) in a competitive fashion, forming contacts that are mutually

exclusive on single alleles. Increasingly, enhancers, gene promoters and CTCF bound sequences are found engaged in contact networks with multiple other local regulatory sequences. Such contacts between individual regulatory sequences and genes, only to be discovered by targeted high-resolution 3C approaches and in very deeply sequenced Hi-C datasets, are functionally the most meaningful chromatin interactions with the potential to directly impact gene expression. They usually take place within the context of topologically associated domains (TADs), chromosomal regions of on average 1 Mb in size that self-interact and thereby form the structural and functional units of chromosomes¹⁰⁻¹². MC-4C was developed to reach resolutions that allow study of such local functionally important contact networks and enable discerning whether regulatory sequences and genes spatially aggregate in chromatin hubs or, oppositely, compete with each other for contacts. In recent work, we showed that MC-4C can identify enhancer hubs, formed between the individual elements of a super enhancer and we demonstrated by MC-4C that some genes compete while others cooperate with each other for interaction with this enhancer hub. MC-4C further allowed demonstrating that CTCF-bound boundaries of neighboring domains aggregate to form rosette-like chromatin structures in interphase cells. This phenomenon became even more apparent upon depletion of the cohesin-unloading factor WAPL, which made us propose that the boundary bouquets we observed are the consequence of ‘traffic jams’ of extruding cohesin molecules that collide into each other if not adequately removed by WAPL¹³.

Applications of MC-4C

MC-4C can be applied to any genomic locus in any cell type or species, where it can uncover how local individual regulatory sequences and genes within a 120 kb to maximally 2 Mb genomic interval (explained below) spatially coordinate their action at the single allele level. The uncovering of these higher-order topological frameworks is expected to facilitate interpreting genotype-phenotype relationships. It can help, for example, to understand why certain regulatory sequences appear redundant while others seem crucial for proper gene expression. Besides regulatory interactions and hubs, we expect MC-4C to be instrumental for the uncovering of periodicity of chromatin folding, if existing in given genomic intervals and at given moments of the cell cycle. Finally, data generated by MC-4C should be extremely useful for the validation and optimization of current 3D genome modeling algorithms^{14,15} that all rely on pairwise proximity ligation measurements to infer the higher-order topologies that MC-4C directly measures.

Comparison with other methods

Recently, several breakthrough methodologies were developed to study multi-way chromatin contacts. One example is C-Walks (chromosomal walks) that employs serial dilution, fragmentation and barcoding of large proximity-ligation concatemers, followed by sequencing and in silico reconstruction of the ligation products¹⁶. C-walks is an untargeted, genome-wide, method that randomly collects and identifies multi-way contacts from across the genome. It was used to investigate the spatial aggregation of intra- and inter-chromosomal regions inside so-called nuclear compartments and therefore essentially analyzes how TADs position themselves relative to each other in the cell nucleus. It was suggested that at this level of spatial organization, preferred multi-way chromatin clustering is rare but may occur for example at polycomb bodies¹⁶. Three-way Hi-C contact analysis methods¹⁷, GAM (genome architecture mapping)¹⁸ and SPRITE (split-pool recognition of interactions by tag extension)¹⁹ are other novel untargeted, genome-wide methods for investigating chromatin clustering at the level of nuclear compartments. SPRITE uses a sophisticated barcoding system based on repeated pooling and splitting of crosslinked DNA ‘hairballs’ to uncover clustered DNA fragments from across the genomes of many cells and enabled uncovering hubs of inter-chromosomal contacts at nucleoli and nuclear speckles¹⁹. GAM is another orthogonal method that is not dependent on DNA digestion and ligation but that isolates and sequences the DNA content of many ultra-thin cryo-fixed nuclear slices for in silico analysis of favored spatial chromosome configurations. GAM also showed preferred multi-way clustering of intra- and inter-chromosomal regions dense in active genes and super enhancers¹⁸.

While very elegant and insightful, none of these methods (currently) provides sufficient local coverage to study the important regulatory contact networks and intricate higher-order topologies that are formed inside TADs. A recently developed targeted multi-contact strategy, called Tri-C²⁰, was designed to study this fine-scale level of 3D genome organisation and was shown to generate data and results similar to that obtained by MC-4C. Tri-C uses biotinylated capture probes instead of inverse PCR methods to enrich for multi-way contacts centered on a viewpoint fragment of interest, in a protocol that requires minimal PCR amplification. For efficient pull-down, the long proximity ligation products are first sheared to ~450 bp products prior to pulldown and Illumina paired-end sequencing is used to analyse the products (~50% contains multiple ligation junctions). The unique ends that are created by sonication serve as unique molecule identifiers (UMIs) for filtering out PCR duplicates. Tri-C can be scaled to analyse millions of multi-way contacts, of which the great majority will be 3-way. In comparison, with MC-4C we typically analyze roughly ten thousand different micro-topologies per single Nanopore

MinION run. Since these are larger, more complex, ligation products, this adds up to analyzing nearly hundred thousand different local contacts per experiment. A distinct feature of MC-4C is the analysis package that comes with the protocol, which we developed to enable robust interpretation of three-way interactions. Recognizing that pairwise contact frequencies in a region of interest are unique for each cell-type of interest and must be considered to interpret multi-way contact frequencies, we designed a statistical method that does not compare multi-way contact profiles from different tissues, but evaluates each dataset independently. Based on a data intrinsic statistical comparison of observed over a thousand background-predicted multi-contact frequencies, z-scores are assigned to co-occur frequencies. This enables robust distinction between cooperative, random and competitive multi-way interactions between regulatory sequences and genes in a locus of interest, as we have shown recently¹³.

Experimental Design

MC-4C is based on 4C technology, a ‘one-versus-all’ contact analysis method that we previously developed to search the genome in an unbiased fashion for sequences contacting a selected DNA fragment of interest^{5,21}. Both methods are based on 3C¹ and involve circularization of proximity ligation products followed by inverse PCR with primers specific for the target fragment of interest, to amplify and analyze the DNA fragments its alleles were ligated to in the cell population. Where the original 4C protocol aims to minimize the sizes of circles that need to be amplified by PCR, MC-4C is modified such that it aims to best preserve the long concatemers with multiple DNA fragments ligated to the fragment of interest. To generate and analyse such long PCR-amplified concatemers, the procedure was tailored as follows and combined with long-read single molecule sequencing on third-generation sequencing platforms like Oxford Nanopore Sequencing or Pacific Biosciences.

As in most 4C and Hi-C protocols, four-cutter restriction enzymes were used for in situ digestion of formaldehyde crosslinked DNA. This results in small DNA fragments (average size 250 bp) that, as a consequence of formaldehyde fixation, often stick together in ‘hairballs’ with multiple other crosslinked DNA fragments. Subsequent in situ ligation fuses these ‘hairball’ DNA fragments together, yielding large concatemers of ligated fragments. We further process these large DNA molecules with a secondary six-cutter restriction enzyme to obtain concatemers with sticky-ends (for efficient ligation-mediated circularization) and of a size (average 4 kb) that will harbor many ligation partners but can still be amplified by PCR with inversely oriented primers that bind to the restriction fragment of interest (the viewpoint). It is essential that this viewpoint-fragment is i) of a sufficient size to accommodate two primers and a guide RNA (gRNA) binding

site in-between the primers; usually between 200 and 500bp; ii) is overlapping or at least in close (<1Kb) linear proximity to the genomic element for which 3-way interactions are to be studied (i.e. overlapping with a promoter, enhancer or bound CTCF site) and iii) does not contain any internal secondary restriction (i.e. HindIII) sites. For the viewpoint fragment, a set of inverse primers must be developed that bind uniquely and are positioned >50bp from the DpnII restriction sites. Allowing some spacing between the restriction sites and the primer binding sites facilitates viewpoint detection after long-read sequencing. We recommend testing the PCR primers on a small batch of MC-4C template, to assure that they yield products in the 2-5 Kb range. If this is not the case, their efficiency can be optimized with alternative annealing temperature and the addition of DMSO and/or MgCl₂ to the PCR reaction, although satisfying results are often faster obtained by simply ordering a new set of primers elsewhere on the same viewpoint. PCR amplification is performed using a highly processive polymerase in a reaction with an extension time that enables replication of the long concatemers. In such reaction, rolling-circle PCRs prioritize amplification of shorter DNA circles. This, we noticed, causes an overrepresentation of small products having only the target fragment of interest with or without its natural neighbor fragments on the linear chromosome. They account for the majority of reads but are not informative for genome topology studies. To eliminate them and target sequencing to informative concatemers, we included an *in-vitro* Cas9 digestion step prior to PCR that simultaneously cuts the target fragment of interest (in between the inversely oriented PCR primer binding sites) and its two neighbor fragments (see Fig. 1b and c). gRNAs can be designed using several available online tools (described in the “**guide RNA design and validation**” section). We routinely first test their effectiveness on genomic DNA or a small amount of MC-4C template prior to their use on bulk MC-4C template. For this, we design a PCR across the Cas9 cleavage target site and measure the *in vitro* digestion efficiency of Cas9 at this site by quantitative PCR (qPCR).

Prior to sequencing, products are size selected and sequenced on a long-read single molecule sequencing platform.

To process, analyze and interpret this new type of sequencing data, a custom pipeline was designed that stringently removes PCR duplicates and is able to identify and quantify multi-way interaction profiles. Importantly, we have developed an analysis pipeline that compares observed three-way interaction frequencies to data-predicted background contact frequencies, allowing data-intrinsic statistical analysis of complex interaction profiles.

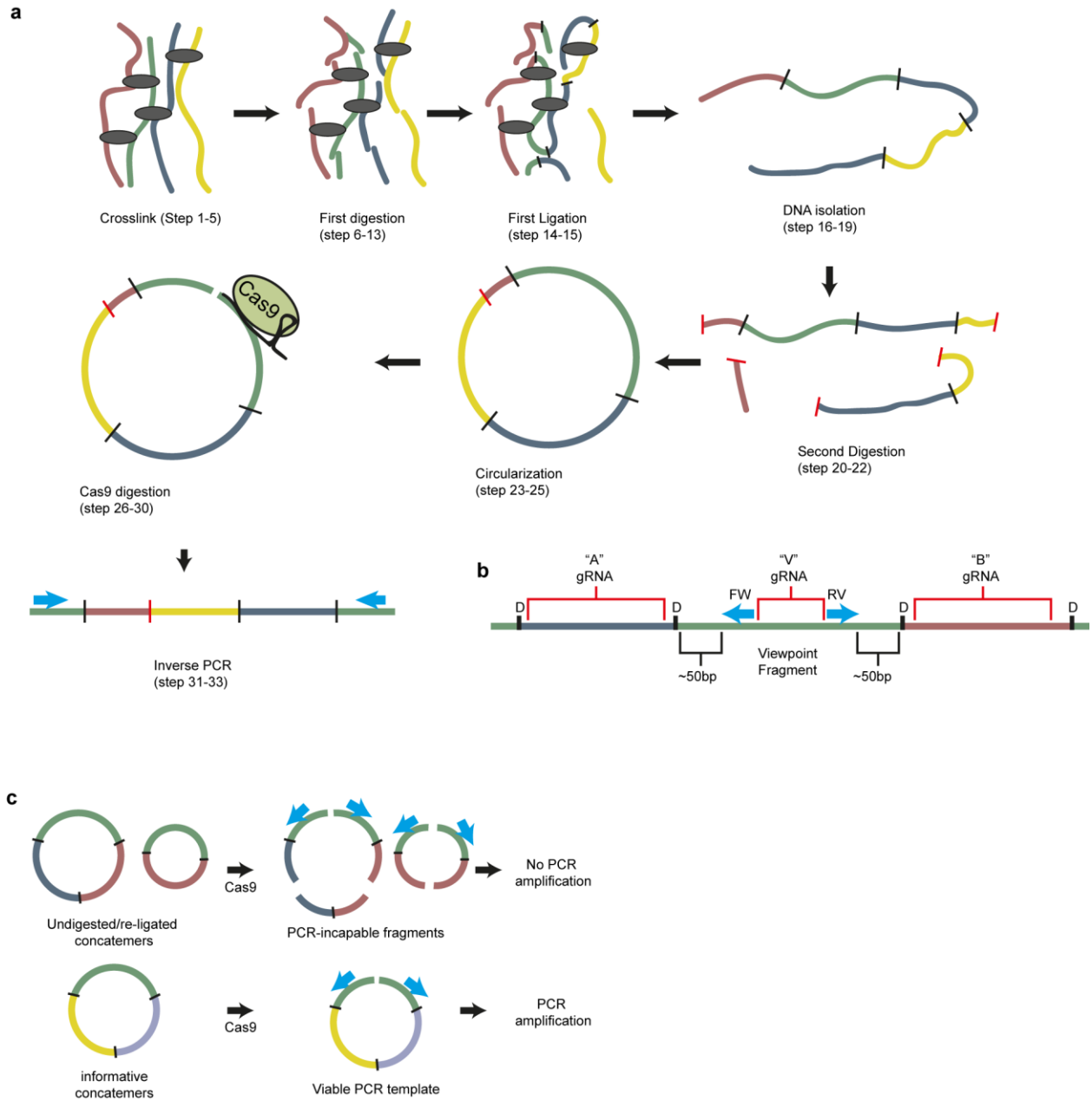


Fig. 1: schematic overview. *a.* Summary of the the MC-4C protocol: chromatin is first crosslinked using formaldehyde, and then digested using a restriction enzyme (DpnII). Subsequently, fragments in close physical proximity are ligated in situ. The DNA concatemers are then isolated and re-digested with a 6-cutter restriction enzyme. Subsequent ligation results in circularization of the template concatemers. A viewpoint-specific Cas9 digestion is then applied to avoid amplification of frequent background products and rollin-circle PCR's. Next a viewpoint-specific inverse PCR is performed. *b.* Primer and gRNA design overview. For a selected viewpoint fragment, a set of inverse primers is designed with approximately 50bp in-between the primer binding site and the

first digestion sites. Three gRNA's are then designed to cut in-between the inverse primers, and in each of the "neighbouring" fragments. c. Rationale of cutting the "neighbouring" fragments. A frequent by-product of proximity ligations are viewpoint-fragment self-ligation, and undigested or re-ligated VP-neighbour fragments. These take up a large proportion of the PCR product and sequencing capacity. When the three gRNA's are successfully applied, such by-products are fragmented, and can no longer yield PCR product, while informative ligations between the viewpoint and non-neighboring fragments are left intact.

Expertise needed to implement the protocol

The preparation of MC-4C template is very similar to our published protocols for the preparation of 4C template^{6,21}, but has been optimized by the use of magnetic beads for DNA purification. Thus, the use of (commercially available or homemade) magnetic separators is required. The final steps of the protocol, size selection and long read single molecule sequencing, require a specialized gel-separating device and an Oxford Nanopore or Pacific Biosciences sequencing device.

Limitations

A limitation of the MC-4C protocol is the high numbers of cells that are required. To accommodate for the loss of material that occurs at various steps of the protocol, and for the fact that the majority of the PCR product is below the ideal >1.5Kb size range, we generally pool 96 individual PCR reactions for each viewpoint, with a total of 100ng of template per PCR reaction. Thus, about 20ug of template is required per viewpoint (where conventional 4C only requires 800ng). Out of 2×10^7 cells we routinely generate 60ug of template, allowing for 3 MC-4C analyses. For most tissue culture cells obtaining 2×10^7 cells is not difficult, but for certain primary tissues or FACS sorted cell populations this may be prohibitive amounts of material. Further, MC-4C only provides meaningful insights into higher order topologies for genomic intervals within a given size range. At the lower size limit we recommend not to consider regions smaller than 120kb: our current analysis strategy pools restriction fragments into a standard number of 200 bins across the region of interest, no matter the size of this region. Below 120 kb the sizes of these bins become smaller than 600bp, i.e. often erroneously implying a resolution below the restriction fragment size. At the upper size limit relevant data interpretation requires a minimum amount of multi-way contacts with distal sites, which relies on

sequence depth. In practice, this limits analysis to contacts within TADs, or between a few adjacent TADs within 1-2 Mb (see further below).

We further notice that, as compared to Illumina sequencing, the sequencing throughput provided by Nanopore and PacBio platforms is still limited (but improving rapidly), and indexing strategies are not yet routine practice. As a result, we currently only run single MC-4C experiments per flow-cell. When different viewpoints are used, the primer sequence should be traceable though and can serve as an inherent barcode, allowing pooling of multiple samples in a single run.

Reagents

FBS	Sigma Aldrich F7524	
Formaldehyde	Sigma Aldrich 1.03999.1000	<i>Very Hazardous, observe MSDS, use in fume-hood and dispose in appropriate waste container</i>
Glycine	J.T.Baker 0582-05	
Tris pH 7.5	Invitrogen 15504-020	
NaCl	Boom 76028327-5000	
Ethylenediaminetetraacetic acid (EDTA)	Invitrogen 15576028	
NP-40	Sigma Aldrich 74385	
Triton	Sigma Aldrich T8787	
DpnII buffer	New England biolabs B05435	
Sodium Dodecyl Sulphate (SDS)	MP biomedical 811030	<i>Precipitates when cooled below room temperature.</i>
DpnII enzyme	New England Biolabs R0543	
Proteinase K	Roche 3115801001	
Ligation buffer	Roche 11243292001	
Ligase	Roche 10799009001	
2-Propanol	Boom 76025379	<i>Flammable</i>

Ethanol	Boom 84028185	<i>Flammable</i>
NEB buffer 2.1	New England Biolabs B72025	
HindIII	New England Biolabs R0104	
Qubit DNA Broad Range	Invitrogen Q32853	
Qubit RNA Broad Range	Invitrogen Q10211	<i>Take care to avoid RNase contamination</i>
Nucleomag 96 PCR Beads (P-Beads)	Macherey Nagel 744100.34	
Cas9	New England Biolabs M0386	<i>Take care to avoid RNase contamination</i>
Cas9 buffer (NEB3.1)	New England Biolabs B7203.5	<i>Take care to avoid RNase contamination</i>
AMPure-XP beads	Beckman Coulter A63881	
PhireII polymerase	Thermo Scientific F122	
DNTP's	Roche 3622614001	
Q5 high fidelity polymerase	New England Biolabs M0491	
T7 Megashortscript kit	Invitrogen AM1354	<i>NTP's do not tolerate freeze-thawing well, we recommend aliquoting the NTPs</i>
Faststart universal Sybr green mastermix mix	Roche 4913914001	<i>Take care to avoid RNase contamination</i>
Equipment		
Eppendorf centrifuge	Eppendorf 5424R	
Swing-out centrifuge	Eppendorf 5810R	
PCR machine	Biorad S1000	
Water bath	Grand JBAqua 5	
Heat block	Eppendorf thermomixer Compact/F1.5	

qPCR machine	Bio-Rad CFX connect
Automated Size selection	Pippin with HTC1510 cassettes
Nanopore sequencing	Oxford Nanopore MinION/GridION
DNA quantifier	Invitrogen Qubit 2.0

Main procedure

Harvesting Cells

For the generation of high-quality template it is essential to obtain a single-cell suspension, resulting in homogenous crosslinking. Cells can be isolated from tissue (see A for an example) or from cultured cells (see B for an example), the quantities specified in this protocol are for 2×10^7 cells. **Troubleshooting: If obtaining a single-cell suspension proves difficult, see Troubleshooting 1.**

A) Harvesting mouse fetal liver cells

- (i) Harvest embryos at E14.5 and remove the dark red liver, remove all lightly colored tissue
- (ii) Gather the livers from the entire litter in 10 mL PBS and keep on ice.
- (iii) Use a P1000 pipet to dissociate livers by pipetting up and down. Place a 40 μ m cell strainer (Greiner bio-one cat. Nr 542040) on a 50 mL falcon tube, and transfer dissociated liver suspension onto the membrane. Pipet the livers up and down while the suspension is kept on the membrane, single cells will pass through the membrane. If large chunks remain, use the plunger of a 5mL syringe to press the chunks through the membrane.
- (iv) Spin down the single cell suspension at 500g for 10 minutes at 4°C in a swing-out centrifuge, discard supernatant and resuspend in 5mL 10%FBS/PBS. Count cells using a counting window or a cell-counter system and proceed with 2×10^7 cells (per batch).

B) Harvesting K562 cells for MC-4C

- (i) Grow cells in suspension, count cells and pool $\sim 2 \times 10^7$ cells into a 50 mL falcon tube (per batch).
- (ii) centrifuge cells at 500g for 10 minutes at room temperature in a swing-out centrifuge, discard supernatant and take up cells in 5 mL 10% FBS/PBS.

Template preparation

Crosslinking and first digestion (day 1, 3 hours hands-on)

- 1) Crosslink the single-cell suspension by adding 5 mL of freshly prepared 4% formaldehyde in 10% FBS/PBS and incubating 10 minutes at room temperature on a rollerbank. *Critical step: making a 4% formaldehyde solution ensures an even distribution of formaldehyde upon mixing compared to adding 37% formaldehyde directly to the cell suspension.* **CAUTION: Formaldehyde is toxic, avoid skin contact, use in a fume-hood and dispose of waste appropriately.**
- 2) Quench the crosslinking reaction by adding 1.425 mL of a 1 M glycine solution and cooling the samples to 4°C. Spin down 10 minutes at 500g in a swing-out centrifuge at 4°C.
- 3) Remove the supernatant and take up the cells in 1 mL cold and freshly prepared lysis buffer. Transfer cells to a 2 mL Eppendorf tube. Incubate for 15 minutes on ice. **Troubleshooting: For some cell types, longer incubation is required for digestion to be effective. This is likely due to a more sturdy cell membrane.**

Lysis buffer		
Component	Final concentration	10 mL example
Tris pH 7.5	50 mM	500 μ L 1M tris Ph 7.5
NaCl	150 mM	300 μ L 5M NaCl
EDTA	5 mM	100 μ L 0.5M EDTA
NP-40	0.5%	250 μ L 20% NP-40
Triton	0.1%	100 μ L 10% Triton
MilliQ water	-	8.75 mL

- 4) Centrifuge for 10 minutes at 500 g at 4°C in a swing-out centrifuge, remove supernatant and resuspend cells in 500 µL 1.1x DpnII buffer. *Note: Cover a 15 mL falcon tube with Kimwipes or other paper tissues to make a centrifuge adapter for 2 mL tubes, allowing the use of a regular swing-out centrifuge. Using a swing-out centrifuge yields more convenient pellets and limits cell loss. Storepoint, most cells can safely be frozen after lysis.*
- 5) Spin down cells for 10 minutes at 500 g at 4°C in a swing-out centrifuge, remove supernatant and resuspend in 500 µL 1.1x DpnII buffer.
- 6) Warm up sample to 37 °C in a heat-block, and then add 15 µL 10% SDS, incubate 1h at 37 °C while shaking at 600 rpm*. *Caution: adding SDS to a cold sample causes it to precipitate. *The shaking intensity required to prevent cell sedimentation varies per cell type.*
- 7) Add 75 µL 20% Triton and incubate 1h at 37°C in a heat-block while shaking at 600 RPM.
- 8) Take 10 µL of the sample and store as undigested control (“U”).
- 9) Add 200 units DpnII restriction enzyme, incubate ~3h at 37°C while shaking at 600 RPM. *Note: we see improved performance if DpnII is added in two multiple batches with some hours in between, the indicated 3 hours is intended as a guideline.*
- 10) Add 200 units DpnII restriction enzyme, incubate overnight at 37 °C while shaking at 600 RPM.

Digestion control and first ligation (day 2, 1 hour hands-on)

- 11) Take 10 µL of the sample and store as digestion control (“D”).
- 12) To both the undigested and digested controls, add 80 µL 10mM Tris Ph 7.5 and 10 µL 10 µg/µL Proteinase K, incubate 1h at 65 °C, then run 20 µL on a 0.6% agarose gel. In case of non-mouse cells, the digested sample should show a smear of DNA with very little or no material above the 10Kb range. In case of mouse cells, a distinct high-molecular weight band always remains in the sample (due to the absence of DpnII sites in the mouse pericentromeric repeats). See **Fig. 2** for examples and see **Troubleshooting** for further steps if the digestion is not satisfactory. *Note: High molecular weight genomic DNA in the undigested control can be difficult to pipet and may rapidly diffuse out of the gel slot, in such cases one can only interpret the digestion efficiency by comparison to the marker.*

- 13) If digestion is as expected, inactivate the restriction enzyme by incubating the sample at 65 °C for 20 minutes while shaking at 600 RPM.
- 14) Add 1200 µL MilliQ and 200 µL, 10x ligation buffer, mix well and then add 10 µL T4 ligase (10u/µL). Incubate overnight at 16 °C. *Note: we achieve this temperature by placing a water bath in a 4 °C room and setting the heat to 16 °C.*

Ligation control and reverse crosslinking (day 3, 1 hour hands-on)

- 15) Take a 30 µL ligation control (“L”) and add 60 µL 10 mM Tris pH 7.5 and 10 µL proteinase K (10 ug/µL), incubate 1h at 65 °C while shaking at 600 RPM and run on agarose gel with the previously obtained digestion control for comparison. A clear upwards shift in molecular weight should be visible. See **Fig. 2b** for expected results and see **Troubleshooting 3** if the ligation is not sufficient (ie. no clear upwards shift is visible on agarose gel).
- 16) If the ligation was successful, add 30 µL of 10 µg/µL proteinase K to the cell suspension and incubate overnight at 65 °C while shaking at 600 RPM.

DNA isolation, second digestion and second ligation (day 4, 3 hours hands-on)

- 17) Split the DNA solution over two 2 mL tubes and add 1 mL of 2-propanol to each tube, mix well until the precipitating DNA becomes visible, incubate 30 minutes at -20 °C.
- 18) Spin down at 12,000 g in a tabletop centrifuge at 4 °C to pellet the DNA, remove supernatant and wash 2x with 1 mL 80% ethanol. Remove as much ethanol as is possible without disturbing the pellet and then air-dry the pellet for 5-10 minutes. *Note: take care not to overdry the DNA pellet.*
- 19) Resuspend each pellet in 180 µL milliQ, incubate for 10 minutes at 50 °C and pool both samples into one 360 µL sample.

Storepoint: resuspend in TE for long term storage of 3C template

- 20) add 40 µL 10x restriction buffer (NEB 2.1) and 50 units HindIII. Incubate 3 hours or more at 37 degrees, put 5 µL of the sample (“D2”) on a 0.6% agarose gel to check the digestion. See **Fig. 2b** for an example.
- 22) If the digestion is satisfactory (a clear downward shift of the smear is observable), heat-inactivate the enzyme by incubating at 65 °C for 20 minutes. Otherwise add 50 more units of HindIII.
- 23) Use a Qubit spectrometer to quantify the amount of template.

- 24) Set up a ligation reaction with a final DNA concentration of 5ng/ μ L; for example, to a batch of 50ug template, add a mix of 1 mL 10x ligation buffer, 8.6mL milliQ and 20 μ L T4 ligase. Incubate overnight at 16 °C.

Template purification and cas9 digestion (Day 5, 2 hours hands-on)

perform a Nucleomag PCR bead (p-bead) purification:

- 25) add 1 sample volume isopropanol
- 26) add 1:100 sample volume Nucleomag PCR beads
- 27) incubate 30 minutes on a rollerbank
- 28) put the sample on a magnetic separator until the solution becomes clear
- 29) remove the supernatant, add 20 mL 80% ethanol, resuspend beads and put the tube back on the magnet.
- 30) wait until all beads are cleared out of the solution, remove supernatant, and wash again with 80% ethanol, resuspend the beads thoroughly and then put the tube back on the magnet.
- 31) wait until all beads are cleared out of the solution, remove all but ~4mL of the supernatant while keeping the tube on the magnet.
- 32) Remove the tube from the magnet, resuspend the beads in the remaining supernatant, transfer 2 mL ethanol/bead suspension to a 2 mL Eppendorf tube and transfer the 2 mL eppendorf tube to a magnetic separator.
- 33) remove and discard the ethanol when the solution becomes clear
- 34) add the remaining 2 mL bead-solution to the 2 mL tube, and wait until the solution clears again.
- 35) Take 1 mL of cleared supernatant, and use it to rinse the remaining beads out of the 50 mL falcon tube, transfer the remaining beads to the 2 mL tube on the magnet rack. (repeat this step if beads still remain in the falcon tube).
- 36) discard the residual ethanol once the solution clears
- 37) Shortly spin down the beads on a tabletop centrifuge or at 1000g for 10 seconds in a microcentrifuge. Transfer the tubes back to the magnet, and remove the residual ethanol. Briefly air-dry (avoid over-drying).
- 38) resuspend the beads in 500 μ L 10 mM Tris pH 7.5/in RNase free water, incubate 10 minutes at 50 °C while shaking at 700 RPM on a heat-block.
- 39) place the tube back on the magnet rack, and transfer the supernatant to a new 1.5 mL Eppendorf tube.

- 40) When residual beads are visible after the first transfer, keep the tube on the magnet rack for 10 minutes and re-transfer the solution to a new tube. Repeat this step until no more beads are visible when removing the sample.

In vitro Cas9 digestion

Caution: Cas9 digestion is viewpoint specific, and dependent on gRNAs. We highly recommend testing gRNAs in a small scale on genomic DNA prior to use on MC-4C template. See “**Guide RNA design and validation**” for instructions on preparing and testing gRNAs.

Caution: Work RNase free from this point onwards.

- 40) For each of the three gRNAs prepare a Cas9-gRNA pre-incubation according to the following recipe (for 20 μg 's of template) and incubate 30 minutes at room temperature:

ingredient	quantity
10x NEB 3.1	10 μL
gRNA (100ng/uL)	10 μL
Cas9 enzyme (1uM)	5 μL
DEPC	75 μL

- 41) Buffer the MC-4C template to be digested, to 500 μL containing 20 μg template add 55 μL 10x NEB3.1

Note: When an efficiency test for the Cas9 digestion is to be performed, some template should be left undigested as a control.

- 42) Add each of the three pre-incubations to the MC-4C template and incubate at 37°C for 3 hours.
- 43) Inactivate Cas9 by adding 1:25th volume of 10% SDS and incubating at 70°C for 10 minutes.

- 44) *Perform a 0.6x AMPure XP cleanup:*
- i) Add 0.6X the total volume of AMPure XP beads to the sample, mix thoroughly by vortexing and incubate 5 minutes at room temperature
 - ii) Place on a magnetic separator and wait until the solution becomes clear
 - iii) Remove the supernatant and add 500 μ L 80% ethanol, do not remove the tube from the magnet or mix
 - iv) Wait until the solution becomes clear
 - v) Repeat step III and IV
 - vi) Remove the supernatant and shortly centrifuge the beads on a tabletop centrifuge (10 seconds at 1000g), place the tube back on the magnet rack.
 - vii) When the beads have moved from the bottom of the tube, remove the residual ethanol.
 - viii) Resuspend the beads in 500 μ L 10 mM tris pH 7.5, incubate 10 minutes at room temperature while shaking at 600 RPM. *note: We recommend elution at 50 °C when the yield is critical (ie. with precious template).*
 - ix) Shortly centrifuge the bead solution (10 seconds at 1000g) and place back on the magnet rack
 - x) When the solution is clear, take the supernatant.
- Optional: Check Cas9 digestion efficiency by qPCR, quantify the template using a qubit spectrophotometer, and perform a qPCR on 30 ng's of template. Compare the CT values between primer sets flanking the cas9 digestion site and a fourth, unrelated primerset (that does not span a DpnII or HindIII restriction site), for Cas9-digested and Cas9-undigested template.*

PCR reaction and purification (Day 6: 2 hours hands-on)

Viewpoint specific PCR

- 45) Perform 96 viewpoint specific PCRs using PhireII hotstart polymerase. And the viewpoint specific primers designed according to the guidelines in “Experimental design”. We usually make a pre-mix containing 5 μ M forward primer and 5 μ M reverse primer.

	1x	96x
5x phire buffer	5 μ L	480 μ L
10 mM DNTP	0.5 μ L	48 μ L
5uM primer mix	2.5 μ L	240 μ L
Polymerase	0.5 μ L	48 μ L
Template	100 ng	9.6 ug
MQ	Up to 25 μ L	Up to 2.4 mL

Run the following program:

98 for 20”

30x

98 for 10”

57 for 20”

72 for 1’30”

72 for 5’

- 46) Pool all 96 PCR reactions, mix well and take a 500 μ L aliquot for size selection and sequencing

- 47) Perform a 0.6x AMPure XP purification on the 500 μ L aliquot (see step 30 of the main procedure), elute in 75uL TE.

- 48) Analyse 20 μ L of the purified PCR product on a 0.6% agarose gel. The result should be a smear that includes fragments above 1.5Kb, and that is not dominated by strong “bands” (see **Fig. 2d**).

Size selection and sequencing (day 7, 2 hours hands-on)

Perform a pippin size selection according to manufacturer's instructions.

We routinely used the Pippin HT HTC1510 machine with a size-selection range between 1.5 and 8 Kb on a 1.5% cassette.

The input DNA can be spread over multiple “lanes”.

Perform a Nanopore library prep according to the manufacturer’s instructions for the 1D genomic DNA by ligation protocol (SQK-LSK109) but:

- Omit shearing the DNA
- Adjust the volume for a higher input quantity when multiple pippin lanes were used.

Guide-RNA design and validation

- 1) For each MC-4C viewpoint, we recommend 3 different gRNAs, one in-between the inverse PCR-primers, and one in each of the two DpnII restriction fragments flanking the viewpoint fragment. We use the ATUM gRNA design tool (<https://www.atum.bio/eCommerce/cas9/input?multipleContacts=false>). Set the design to wild-type Cas9, and enter the restriction fragment sequence into the “Target a sequence” dialog box (In some cases, extremely small DpnII fragments flank a viewpoint, in such cases we target the closest fragment where gRNA design is possible). Select the gRNA with the highest score (it is advisable to perform a BLAST search for off-targets), and click “continue”. This will show the gRNA sequence to be used, excluding the CRISPR sequence.

Note: It is recommended to also design a qPCR primer set within the restriction fragment, that overlaps the Cas9 cut-site so Cas9 digestion efficiency on MC-4C template can be ascertained using qPCR.

- 2) gRNA’s can be obtained through *in vitro* transcription, but in this case a T7 promoter must be included in the template DNA as well as the CRISPR sequence. Therefore we order DNA oligonucleotide consisting of the following sequences for each gRNA sequence obtained in step 1:

5' -AAGCTAATACGACTCACTATAG**G**-gRNA-
GTTTTAGAGCTAGAAATAGCAAG

**note that in vitro transcription initiates at the second G-nucleotide of the GG dinucleotide at the 3' end of the T7 promoter sequence (marked in bold). If the gRNA starts with a G, this last G of the T7 promoter can be omitted.*

3) perform a gRNA template PCR using Q5 high fidelity polymerase, using a universal reverse primer together with the gRNA specific primer to make double stranded template for the *in vitro* transcription:

Universal reverse primer:

5' AAAAGCACCGACTCGGTGCCACTTTTTCAAGTTGATAACGGACTAGCCT
TATTTTAACTTGCTATTTCTAGCTCTAAAC

The FW and universal RV primer can be pooled into a pre-mix containing 5 μ M of each primer.

	1x	4x
5x Q5 Reaction Buffer	10 μ L	40 μ L
5 μ M primer mix	2,5 μ L	10 μ L
10 mM dNTPs	1 μ L	4 μ L
polymerase	0.5 μ L	2 μ L
MQ	36.0 μ L	144 μ L

Run the following program:

98°C for 30 seconds

35x

98°C for 10 seconds

57°C for 10 seconds

72°C for 10 seconds

72°C for 5 minutes

The resulting gRNA template product should be 125 bp. Optionally this be verified on a 2% agarose gel.

- 3) Purify the PCR product using Qiagen PCR purification kit, elute in 25 μL RNase free water, and quantify using the Qubit spectrophotometer and the DNA broad range kit.
- 4) Perform the in vitro transcription using the T7 Megashortscript kit (while taking care to avoid RNase contamination), assemble the reaction in the following order:

DEPC	X (up to 10 μL Reaction volume)
Template DNA (500 ng)	Up to 4 μL
10x Transcription buffer	1 μL
NTP mix	4 μL
T7 enzyme	1 μL

Mix well and incubate at 37 °C for 4 hours

- 5) Add 0.5 μL TURBO DNase to remove template.
- 6) Perform a 4x AMPure XP purification (see step 44 of the main procedure, but add 40 μL AMPure XP beads to the sample) Elute in 100 μL RNase free water, and quantify using the Qubit RNA BR kit. Store at -80 °C. *Optional: gRNA's can be aliquoted if they are to be used repetitively to avoid degradation due to freeze-thaw cycling.*
- 7) *Optional:* Test gRNA efficiency on 5 μg genomic DNA, or if available MC-4C template.

i) Prepare a Cas9 pre-incubation:

ingredient	Quantity
10x NEB 3.1	1,25 μ L
gRNA (100ng/uL)	1,25 μ L
Cas9 enzyme (1uM)	0.6 μ L
DEPC	9,4 μ L

Incubate 30 minutes at room temperature

ii) buffer the gDNA solution containing 5 ug of gDNA with 10x NEB3.1 e.g.,
add 11 μ l 10x NEB3.1 to 100 μ L gDNA solution

iii) add the Cas9 pre-incubation(s) and incubate 3h at 37 °C.

iv) add 1/25th volume of 10% SDS and incubate 10 minutes at 70°C to
inactivate the Cas9 enzyme.

v) purify the template using a 0.6x AMPure XP purification (see step 44 of the
main procedure).

vi) Perform a qPCR using qPCR primers that overlap the Cas9 cleavage site,
while using uncut gDNA or MC-4C template as a negative control.
Efficiency is usually 80-95% (3-6 CT values).

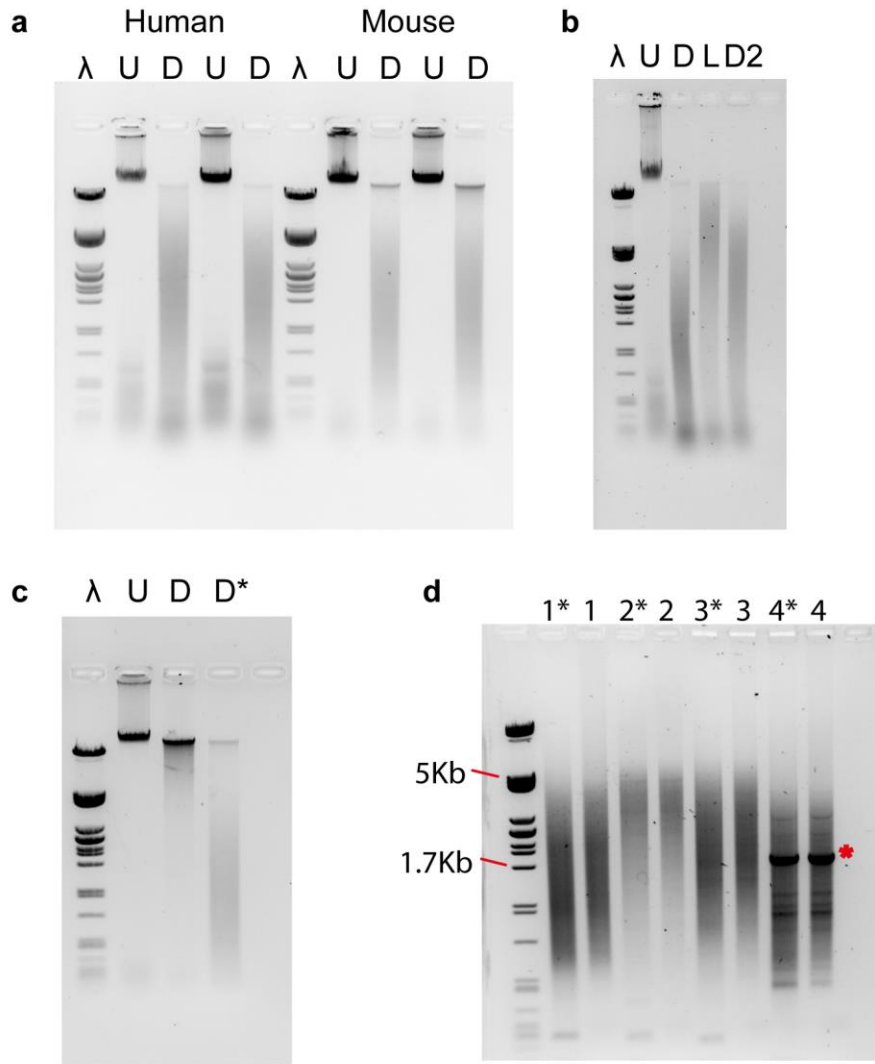


Fig. 2: Intermediate controls in the MC-4C protocol. *a* Examples of DpnII digestions in human and mouse tissue. λ Marks λ-phage DNA digested with PstI restriction enzyme, U marks the lanes with undigested DNA, D marks DpnII digested DNA originating from K562 human cells on the left and mouse embryonic liver cells to the right. Note that DpnII digested mouse DNA always contains a small amount of high-molecular weight DNA, due to the absence of DpnII restriction site in the pericentromere repeats. *b*. An example of all intermediate controls of an MC-4C template from K562 cells. λ Indicated the ladder, U indicated undigested DNA, D indicates the first digestion (DpnII), L indicates the ligation and D2 indicates the second digestion (HindIII). *c*. Example of a case where the first digestion was not adequate, as a large portion of DNA was still >10Kb in size (D). More DpnII enzyme was added and incubation was extended. This resulted in properly digested DNA (D*). *d*. Examples of MC-4C PCR products. These PCRs inevitably vary in size distribution, but should contain material between 1.7 and

5Kb. Each number indicates a different primerset, numbers with an asterisk indicated PCR prior to AMPure size exclusion, and lanes without asterisk contain the final product. Lane 4 is an example of an experiment where the Cas9 digestion partially failed, resulting in a large band (indicated with a red asterix).

Data pre-processing

In this section, we will describe the MC-4C pipeline which is designed to facilitate data processing, analysis and interpretation of sequenced MC-4C experiment. It should be noted that this pipeline is under continuous development which adds new functionalities or enhances the existing ones. Therefore, we recommend the user to refer to the github repository of MC-4C pipeline (https://github.com/aallahyar/mc4c_py) to access the latest version of the MC-4C pipeline as well as the corresponding manual which describes the newly added features or enhancements in the pipeline.

Setting up the MC-4C pipeline:

The MC-4C pipeline requires the following tools:

- A Unix like shell (e.g. Bash v3.2+)
- samtools v1.9+
- bwa v0.7.17+
- Python v2.7+ and the following Python packages:
 - h5py v2.7.1+
 - numpy v1.13.3+
 - pandas v0.23.4+
 - pysam v0.15.1+
 - matplotlib v2.1.2+ (only for producing quality check and analysis plots)

Configuration files:

Experiment specific parameters for each MC-4C dataset are organized in a configuration file. Parameters in this file are defined line by line and stored in a two-column tab-delimited format (i.e. parameter name and value). Multiple values for a parameter are separated by a “,” sign. For example, the following configuration file specifies that the viewpoint lies on chromosome 7, fragments should be mapped to the mm9 mouse reference genome and finally GATC and AAGCTT are used as first and second restriction enzymes to prepare the library:


```
vp_chr <tab> chr7
genome_build <tab> mm9
re_seq <tab> GATC,AAGCTT
```

List of parameters that are recognized in the MC-4C pipeline and can be stored in a configuration file are shown in **Table.1**.

Table.1. Description of parameters that can be defined in a configuration file to be used for processing an MC-4C experiment. Required parameters are denoted by (*).

Name	Value example	Description
genome_build*	hg19	Reference genome of interest
vp_chr*	chr7	Viewpoint chromosome; the chromosome for which the viewpoint primers are designed
prm_start*	110977147,110977593	Start coordinate of primers used. Coordinates are separated by “,”.
prm_end*	110977171,110977623	End coordinate of primers used. Coordinates are separated by “,”.
prm_seq*	GATTTGTGAGCTC AGGGTTTAC,GCAG TAGTGATTCTATTC AATTTTTGGG	Sequence of primers used. Separated by “,”.
re_name	DpnII,HindIII	Restriction enzyme name used to prepare the MC-4C library
re_seq*	GATC,AAGCTT	Restriction enzyme sequence used to prepare the MC-4C library
bwa*	/bin/bwa	Path to BWA aligner.
bwa_index*	/bwa_indices/mm9	Path to corresponding bwa index of reference genome.
reference_fasta*	/genomes/mm9.fa	Path to the corresponding reference genome (in fasta format).
roi_start	110933500	Start position of Region Of Interest (ROI). ROI for example will be used to define ‘far-cis’ fragments in PCR duplicate filter. This parameter will be set to 1Mb up stream of the smallest primer coordinate if not given.
roi_end	111066500	End position of Region Of Interest (ROI). This parameter will be set to 1Mb downstream of the largest primer coordinate if not given.

Note: Selection of a Region of Interest (ROI) requires careful consideration. This is because ROI width has an important role in several parts of the MC-4C data processing pipeline. Initially, the ROI discerns far-cis fragments that can be used as UMIs for duplication removal procedure. Additionally, it identifies “informative” reads (i.e reads with more than one fragments in the ROI) that can be used in multi-contact analysis. Finally, the multi-contact analysis is performed by splitting the ROI into 200 non-overlapping and equally-spaced bins. The user is strongly advised to consider the following suggestions when determining the coordinates for ROI:

- The viewpoint should always be included in the ROI. This is important as exclusion of viewpoint can change the expected distribution of fragments in the ROI which in turn results in unreliable z-scores calculated in the multi-contact analysis.
- If an experiment is expected to investigate interactions within a given TAD, the ROI should encompass the TAD boundaries and a few dozen kilobases of flanking sequences, as done previously ¹³.
- The ROI size should not be smaller than 120Kb when MC-4C library is prepared using DpnII restriction enzyme as the first cutter. Smaller ROIs will result in a bin width less than 600bp (i.e. 120Kb / 200) which is close to expected minimum resolution of such MC-4C experiment (for human or mouse genomes).
- The ROI can be larger if interactions between TAD boundaries are investigated as done previously ¹³. However, considering the expected coverage of a (4bp cutter) MC-4C experiment, the ROI should not be larger than 2Mb as sufficient coverage is required for a reliable multi-contact analysis and further interpretation of results (further discussed below).

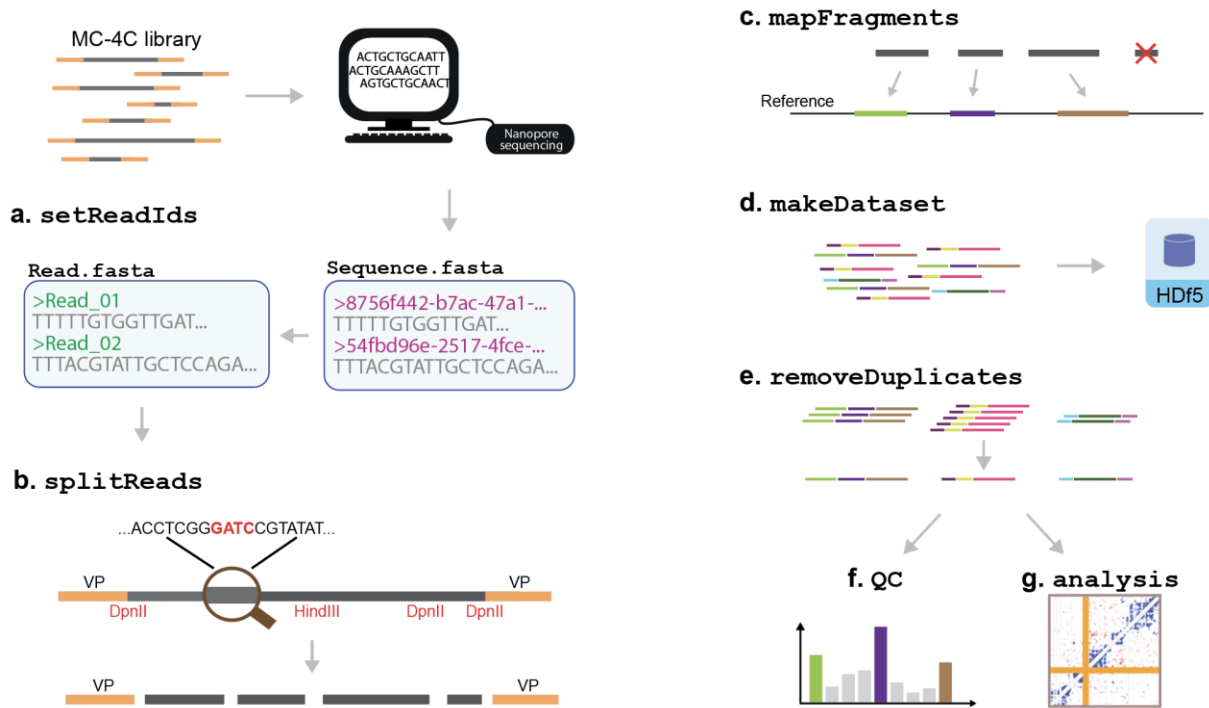


Fig. 3: Schematic overview of the MC-4C pipeline and its modules. Each module is responsible for a specific task. The MC-4C pipeline consists of the following modules: **a.** Identifiers are assigned to each sequenced read to facilitate data tracking across the MC-4C pipeline. **b.** Reads are split into fragments according to restriction enzyme recognition sequences found in each sequenced read. **c.** Fragments are mapped to the reference genome (according to the configuration file). **d.** Mapped fragments are stored in a dataset (in HDF5 format). **e.** Duplicates are removed by discarding all (but one) reads with identical UMI. **f.** The user can use commands in **QC** module to assess quality of performed MC-4C experiment. **g.** Commands in **analysis** module can be used to investigate the multi-contact associations captured in an MC-4C dataset.

Annotations

Several quality control or analysis in the MC-4C pipeline use or require annotations. For example, in the SOI-SOI analysis, each SOI should have coordinates assigned to it to be used in the association test and plotting. For each analysis, only annotations within the ROI are considered. These annotations can be stored in a tab-delimited file which is named according to the genome build of interest (e.g. mm9 or h19). For example, SOIs in the mm9 genome should be stored in an annotation file named:

```
./annotations/ant_mm9.tsv
```

which follows the format below for coordinates:

```
Hbb-bh1 <tab> chr7 <tab> 110990974
```

```
HS1 <tab> chr7 <tab> 111007686
HS2 <tab> chr7 <tab> 111009600
```

Modules:

The MC-4C pipeline is partitioned into modules. Each module performs a specific task such as mapping fragments to the reference genome or removing PCR duplicates from a given MC-4C dataset. List of available modules in the MC-4C pipeline is given in **Table.2**. For example, the user can map the fragments to the reference genome by calling **mapFragments** module using the following command:

```
$ mc4c.py mapFragments <config_file> --input_file <input_file> --
output_file <output_file>
```

In this protocol, the module names are shown in boldface and <config_file>, <input_file>, <output_file> placeholders for the path to the configuration file, input file and output file, respectively. Arguments that are not provided will be obtained from the configuration file if possible (see [MC-4C pipeline repository](#) for further details).

Table.2. Modules defined in MC-4C.

Module name	Function
setReadIds	Defines an identifier for each sequenced read.
splitReads	Splits each read into fragments according to restriction enzyme recognition sequence.
mapFragments	Maps the fragments to the reference genome.
makeDataset	Creates a dataset (in HDF5 format) containing mapped fragments.
removeDuplicates	Removes duplicate reads from an MC-4C dataset.
QC	Generates various summary report plots for an MC-4C dataset.
analysis	Performs multi-contact analysis including VP-SOI (a single SOI vs. all SOIs), SOI-SOI (every pair of SOIs) or Cross-ROI (taking every 3-bins as a SOI) analysis.

Walkthrough of MC-4C pipeline:

In this section, we provide a step by step description of how the MC-4C pipeline can be used to process sequenced reads in an MC-4C experiment. In this walkthrough, we assume that the user is using Linux operating system. Minor modifications might be required in the commands used to follow this walkthrough using Mac OSX.

For demonstration and testing purposes, we prepared a test MC-4C dataset. This dataset consists of a small sequencing file (i.e. raw_BMaj-test.fastq.gz) and a corresponding configuration file (cfg_BMaj-test.cfg) that holds experiment specific details of this experiment. These two files can be downloaded from [here](#).

Setting up the pipeline

- 1) confirm that the required Python packages for MC-4C pipeline are installed by:

```
$ pip install h5py numpy pandas pysam matplotlib
```

- 2) Download the latest version of MC-4C pipeline from its [git repository](#) using:

```
$ wget https://github.com/aallahyar/mc4c_py/archive/master.zip
$ unzip master.zip
$ cd ./mc4c_py-master
```

- 3) Make an index for the reference genome of interest (mm9 in this example) using bwa by:

```
$ bwa index ~/references/mm9.fa
```

Prepare input data:

- 4) After downloading the aforementioned example data, create a folder named “fastqs” and move the obtained sequencing file (i.e. raw_BMaj-test.fastq.gz) to this folder:

```
$ mkdir -p ./fastqs
$ mv ~/Downloads/raw_BMaj-test.fastq.gz ./fastqs/
```

- 5) Create a folder named “configs” and move the obtained configuration file (i.e. cfg_BMaj-test.cfg) to this folder:

```
$ mkdir -p ./configs
$ mv ~/Downloads/cfg_BMaj-test.cfg ./configs/
```

- 6) Define the paths for the reference genome, bwa and the reference index (made in step 3) of the walkthrough) in the corresponding configuration file (i.e. cfg_BMaj-test.cfg). Once done, confirm that these paths are correctly defined by:

```
$ grep 'bwa|bwa_index|reference_fasta' ./configs/cfg_BMaj-test.cfg
```

For example, in our system these paths are defined as follows:

```
bwa /bin/bwa
bwa_index /bwa_index/mm9.fa
reference_fasta /reference_genomes/mm9.fa
```

Assign read identifiers to sequenced reads (**setReadIds**):

In the MC-4C pipeline, aiming to facilitate tracking of reads and fragments that originate from these reads, a unique identifier is assigned to each sequenced read. This is the first step in processing reads produced in an MC-4C library.

7) Assign a unique identifier to each sequenced read using **setReadIds** module by:

```
$ python ./mc4c.py setReadIds BMaj-test
```

This module will produce the following output in the terminal:

```
%% Assigning traceable identifiers to reads ...
Writing reads with traceable identifiers to: ./reads/rd_BMaj-
test.fasta.gz
Total of [1] files are given as input.
  1. Reading from: ./fastqs/raw_BMaj-test.fastq.gz
     processed 5,000 reads.
[i] Read identifier conversion is completed successfully.
```

Split reads into fragments

Each sequenced read in an MC-4C experiment consists of multiple interacting fragments. To assist the aligner in recognizing and mapping these fragments to the reference genome, we opted to pre-split sequenced reads into fragments based on the restriction enzyme recognition sequence.

9) Apply the **splitReads** module by executing the following command:

```
$ python ./mc4c.py splitReads BMaj-test
```

This module will produce the following output in the terminal:

```
%% Splitting reads into fragments ...
Reading reads from: ./reads/rd_BMaj-test.fasta.gz
Writing fragments to: ./fragments/frg_BMaj-test.fasta.gz
     processed 5,000 reads and produced 36,165 fragments.
[i] Total of 5,000 reads and 36,166 fragments are produced
     successfully.
```

Map fragments to the reference genome

10) map the produced fragments to the reference genome using the **mapFragments** module:

```
$ python ./mc4c.py mapFragments BMaj-test
```

The pipeline will produce the following output in the terminal:

```
%% Mapping fragments to genome ...
Reading fragments from: ./fragments/frg_BMaj-test.fasta.gz
Writing mapped fragments to: ./bams/bam_BMaj-test.bam
Running: ~/bin/bwa-0.7.17/bwa bwasw -b 5 -q 2 -r 1 -z 5 -T 15 -t 6
~/references/mm9.fa ./fragments/frg_BMaj-test.fasta.gz | samtools
view -q 1 -hbS - > ./bams/bam_BMaj-test.bam
[i] Fragments are mapped to genome successfully.
```

Making an MC-4C dataset from mapped fragments (makeDataset):

Mapped fragments are computationally extended to the nearest restriction site in the reference genome and adjacent fragments within a read that also map adjacently (<20bp) in the genome are merged.

11) apply the **makeDataset** module:

```
$ python ./mc4c.py makeDataset BMaj-test
```

Once called, the following output will be shown in the terminal:

```
%% Creating an MC-4C dataset from mapped fragments ...
Reading fragments from: ./bams/bam_BMaj-test.bam
Database of restriction enzyme cut sites is not found. Scanning the
reference genome to create this database ...
Searching in reference: ~/references/mm9.fa
Scanning: chr1, chr10, chr11, chr12, chr13, chr14, chr15, chr16,
chr17, chr18, chr19, chr2, chr3, chr4, chr5, chr6, chr7, chr8,
chr9, chrM, chrX, chrY,
Writing processed fragments to a temporary file first:
./datasets/mc4c_BMaj-test_all.hdf5.tmp
processed 0 fragments in 0 reads.
processed 10,000 fragments in 1,640 reads.
processed 20,000 fragments in 3,311 reads.
Loading the temporary file: ./datasets/mc4c_BMaj-test_all.hdf5.tmp
```

```
Writing dataset to: ./datasets/mc4c_BMaj-test_all.hdf5
Removing temporary file: ./datasets/mc4c_BMaj-test_all.hdf5.tmp
[i] MC4C dataset is created successfully.
```

Remove read duplicates (**removeDuplicates**)

Multi-contact data uniquely allows the removal of PCR duplicates based on the collection of ligated fragments. However due to the (currently) limited sequencing quality of third generation sequencers, fragments that are identified in one PCR duplicate may not be identified in another copy. Therefore we opted to use trans-ligations (and far-cis fragments) as genomic UMI's, arguing that the chance of a viewpoint-concatemer randomly ligating to the same trans (or far-cis) fragment multiple times independently is extremely small.

12) To remove PCR duplicate reads in an MC-4C dataset, the **removeDuplicates** module can be called using:

```
$ python ./mc4c.py removeDuplicates BMaj-test
```

This module will produce the following output in the terminal:

```
%% Removing pcr duplicates from an MC-4C dataset ...
Reading MC4C dataset from: ./datasets/mc4c_BMaj-test_all.hdf5
Writing unique MC4C reads to: ./datasets/mc4c_BMaj-test_uniq.hdf5
There are 4990 reads in the dataset.
Selected non-overlapping fragments with MQ >= 20: 4960 reads are
left.
Selected reads #cis fragment > 1: 697 reads are selected.
Selected reads with #trans fragment > 0: 500 reads are selected.
Scanning for duplicated trans-fragments:
  scanned 0 trans-fragments, 500 reads are still unique.
  scanned 500 trans-fragments, 476 reads are still unique.
Result statistics (before --> after filtering):
  #reads: 4,990 --> 475
  #fragments: 20,794 --> 961
Writing dataset to: ./datasets/mc4c_BMaj-test_uniq.hdf5
[i] PCR duplicates are removed from MC4C dataset successfully.
```

This is the final pre-processing stage in the MC-4C pipeline.

Data quality control

The MC-4C pipeline implements comprehensive quality control functionalities in the **QC** module. It is recommended to employ the **QC** module to produce several summary reports that describe general statistics (e.g. read size distribution) for an MC-4C experiment. The full list of commands available in this module is represented in **Table.3**.

Table.3. Commands available in the **QC** module of the MC-4C pipeline.

Command name	Function
readSizeDist	Plots size distribution of sequenced reads in terms of number of base pairs.
cirSizeDist	Plot circle size distribution in terms of number of fragments.
chrCvg	Plots chromosome coverage of mapped fragments.
overallProfile	Plot overall profile of an MC-4C experiment
seqSaturation	Produce statistics to assess sequence saturation and UMIs diversity.

These statistics help the user to assess quality of an MC-4C experiment and also to identify the potential shortcomings in the library preparation that resulted in a suboptimal throughput in an MC-4C experiment.

Read size distribution:

A dataset with many short (i.e. <1.5kb) reads is likely to have insufficient multi-contact complexity (i.e. <2 interactions, excluding the viewpoint). Therefore, it is recommended to initially check size distribution of sequenced reads using the `readSizeDist` command in **QC** module. **Fig. 4** represents read size distributions of two MC-4C datasets from mouse (mm9) and human (hg19). In a successful experiment, we expect a substantial number of reads to fall within 1.5-8kb interval. See **Troubleshooting** for further actions if the distribution of read size in a given experiment is not as expected.

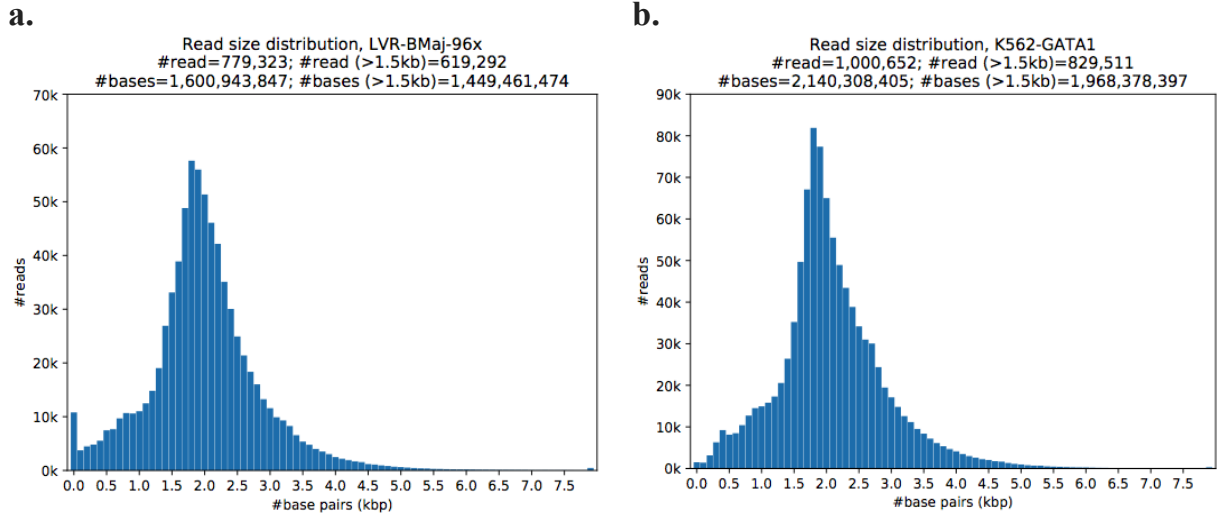


Fig. 4: Read size distribution of two MC-4C datasets. *The top of each plot shows series of characteristics for a given MC-4C dataset: **#read** indicated the total number of sequenced reads, **#read(>1.5Kb)** indicated the number of reads that are larger than 1.5 Kb, **#bases** indicates the total number of sequenced bases, and **#bases(>1.5Kb)** indicated the number of bases in reads over 1.5 Kb.*

Chromosome coverage:

As MC-4C is a targeted approach, it is expected that the majority of fragments to map to the viewpoint chromosome. This is often dictated by specificity of designed primers, their amplification efficiency, or presence of contamination in the produced library. To validate these properties in an MC-4C dataset, chromosome coverages plots can be employed that can be drawn by `chrCvg` command in the **QC** module. **Fig. 5** demonstrates two examples produced by this command.

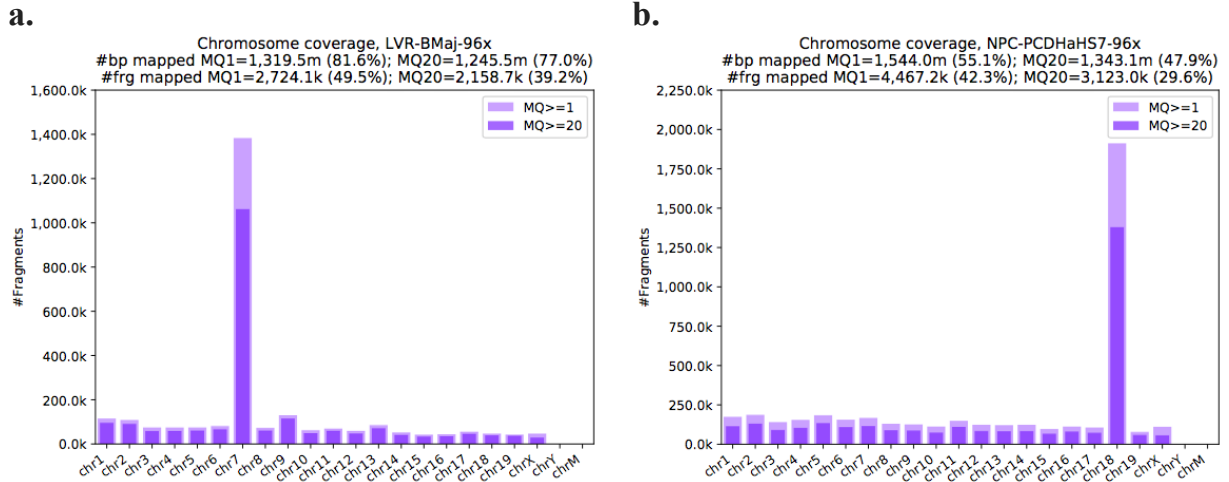
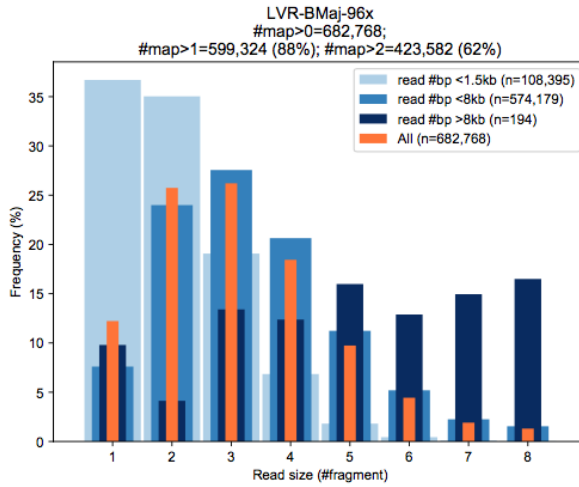


Fig. 5: Chromosome coverage of two example runs. *Light and dark purple bars indicate coverage of mapped fragments with Mapping Quality (MQ) of at least one, and at least 20 respectively. Most fragments are mapped to the viewpoint chromosomes (a. chr7 and b. chr18). On top of the graph, some characteristics are described; respectively the number of bases mapped with a mapping quality of at least one, and of at least 20, and on the bottom line the number of fragments mapped with a mapping quality of at least 1 or at least 20.*

Circle size distribution:

The `cirSizeDist` command enables inspection of the distribution of the number of fragments mapped per read. This is an important measure as multi-way interactions only follow from reads with three or more mapped fragments. The distribution of the number of fragments mapped per read can be calculated using only ROI-fragments (fragments mapped within the ROI), or using only unique reads (duplicates removed). Both are supported in the `cirSizeDist` command through the “`--roi-only`” and “`--uniq-only`” arguments.

a.



b.

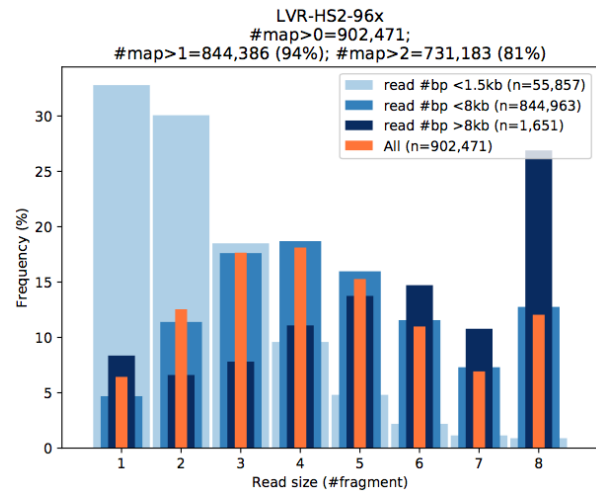


Fig. 7: Circle size distribution of two MC-4C datasets. Each bar represent circles containing number of fragments. Light blue bars represent the number of fragments in reads smaller than 1.5Kb, regular blue represents reads smaller than 8Kb, dark blue represents reads >8Kb. Orange bars indicate the distribution of all reads. The characteristics on top of the graph indicate the number of reads with more than 0 fragments mapped, with more than 1 mapped fragment and the number of reads with more than 2 fragments mapped.

Overall MC-4C profile in the region of interest:

The multi-contact data in an MC-4C dataset can be visualized in a similar fashion to traditional 4C with the `overallProfile` command. An example of such a plot is demonstrated in **Fig. 8**. A suitable overall profile shows coverage across the ROI where coverage reduces on regions that are farther away from the viewpoint.

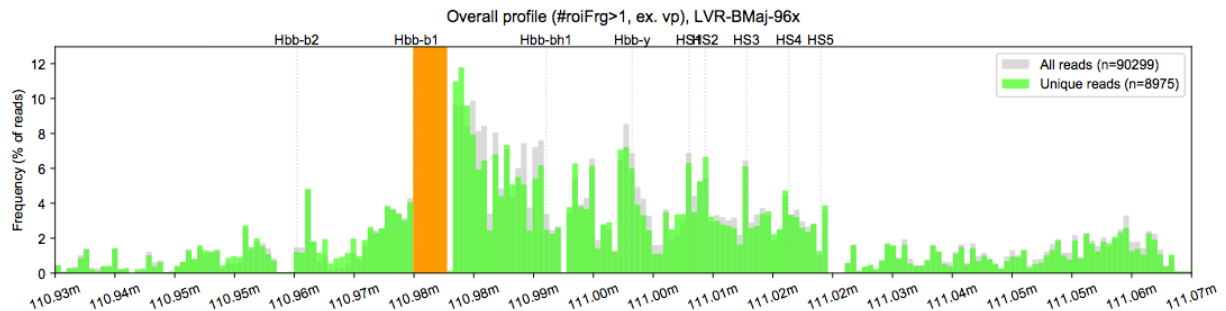


Fig. 8: Overall profile of an MC-4C dataset. Grey bars indicate all mapped reads, and green bars indicate unique reads only. The number of reads is indicated in the legend. Note that the y-axis is plotted in % of total (sequenced or unique) reads.

Sequencing saturation level:

Insufficient complexity (i.e. diversity of reads amplified) in an MC-4C library can result in low yield of unique reads. To assess the obtained complexity of an MC-4C experiment, the `seqSaturation` command in **QC** module can be used. This command produces two panels (see **Fig. 9**). Initially this command investigates the sequencing saturation by downsampling reads in a given MC-4C experiment and plotting the number of reads versus the number of unique reads (see **Fig. 9** left panel). If the plotted line reaches a plateau (ie. sequencing more reads does not yield more unique reads), the user can conclude that the current library is saturated. Additionally, this command calculates a UMI duplicity score (The frequency of an observed UMI divided by the total number of informative reads, multiplied by 1×10^5). By plotting top 100 UMI duplicity scores one can estimate the library complexity. The average duplicity score of all UMIs can be used to the same effect, and is typically below 20. Generally, a MC-4C dataset with sufficient diversity of UMIs should have an average duplicity score below 20. Refer to **Troubleshooting** for further actions if the UMI duplicity score is substantially higher than 20.

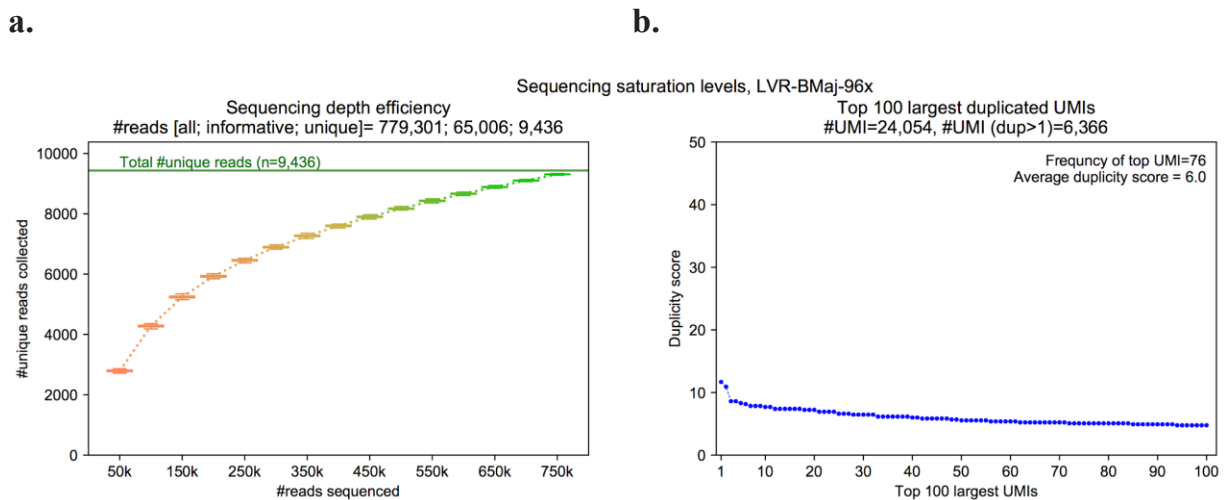


Fig. 9: Quality check for sequencing saturation and UMI diversity. *a.* the number of reads sequenced in the MC-4C experiment is compared to the number of unique reads gained after analysing the downsampled experiment. A sufficiently sequenced library shows a plateau which indicates that not many unique reads will be sequenced when more reads are obtained from that library. *b.* the complexity is investigated by plotting the frequency of top UMIs observed in the MC-4C dataset, represented as a duplicity score. Additionally, average duplicity score of the top 100 UMIs is shown (see top right side of the plot). An average duplicity score below 20 indicates a desired dataset with sufficient complexity. On top of the right panel, the total number of UMI's is shown, as well as the number of UMI's that was sequenced more than once.

Main protocol (7 days)

Day1: Crosslinking and 1st digestion; 3 hours hands-on

Day2: Check digestion, and 1st ligation: 1 hour hands-on

Day3: Check ligation and start reverse crosslinking: 1 hour hands-on

Day4: Purification, second digestion and second ligation: 3 hours hands-on

Day5: Purification, Cas9 digestion: 2 hours hands-on

Day6: PCR reaction and purification: 2 hours hands-on

Day7: Pippin size selection and sequencing: 2 hours hands-on

gRNA preparation (2 days)

Day1: Prepare gRNA template, set up *in vitro* transcription overnight.

Day2: Purify gRNA, set up Cas9 test digestion

Day3: Purify digested DNA and perform qPCR to confirm digestion.

Running MC-4C pipeline (3 hours)

- **setReadIds** module (30 minutes)
- **splitReads** module (45 minutes)
- **mapFragments** module (1 hour)
- **makeDataset** module (15 minutes)
- **removeDuplicates** module (10 minutes)

Results

Three types of multi-contact analysis is supported in the MC-4C pipeline. **Table.4** lists the commands that are responsible for performing these analysis.

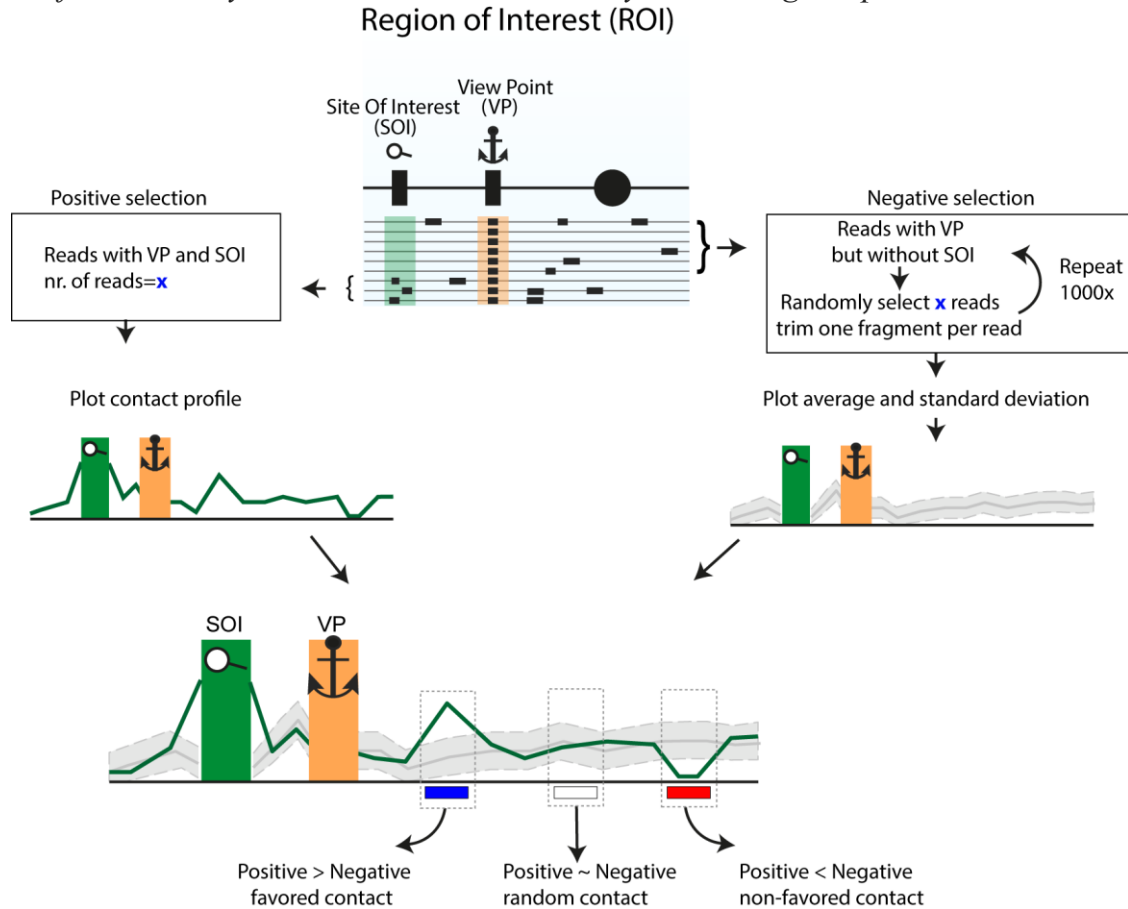
Table.4. Commands available in the **analysis** module of the MC-4C pipeline.

Command name	Function
atVpSoi	Performs multi-contact association test across all bins when a single Site Of Interest (SOI) is interacting with the view point.
atSOISOI	Perform multi-contact association test for each pair of SOI.
atAcrossROI	Performs multi-contact association test across all bins while each three consecutive bins in the ROI is considered as a SOI.

Association analysis of a single site of interest (**atVpSoi**)

This analysis is performed when the user is investigating whether the interaction propensity of the viewpoint and a given bin within the Region Of Interest (ROI) changes when a Site Of Interest (SOI) is present in the same read. This analysis is implemented in **vpSOI** command within the **analysis** module. **Fig. 9.** shows an example plot produced by this command. We note that a robust association analysis for a given SOI requires a minimum number of reads to contain the given SOI. As a guideline, we do not consider the z-scores reliable for any VP-SOI analysis where fewer than 100 reads contained the given SOI.

VP-SOI analysis: This analysis is devised to investigate the three-way associations between view point, SOI and any given third partner in the ROI. The corresponding details for this analysis are described extensively in our original publication¹³.



In summary, we compare the chance of three-way interaction between the viewpoint, the SOI and a given site as the third partner (i.e. the positive set), to a background profile that reflects the three-way interactions of the viewpoint and the same site in absence of the SOI (i.e. the negative set). To estimate the background, we randomly

sample a number of reads from the negative set equal to the number of reads in the positive set. We randomly remove a fragment from each sampled read in the negative set to compensate for the SOI fragment present in all reads in the positive set. This random sampling is repeated 1000x to generate an average background profile and corresponding standard deviation. The VP-SOI contact profile is then compared to the background profile, and a z-score is calculated. When the positive profile is higher than the background, a genomic region is more likely to interact with the viewpoint in the presence of the SOI. A positive profile similar to the background profile indicates that such region is equally likely to interact with the viewpoint with or without presence of the SOI. When the positive profile is lower than the background, a region is less likely to contact the viewpoint when it is interacting with the SOI. *figure adapted from Allahyar et al. ¹³.

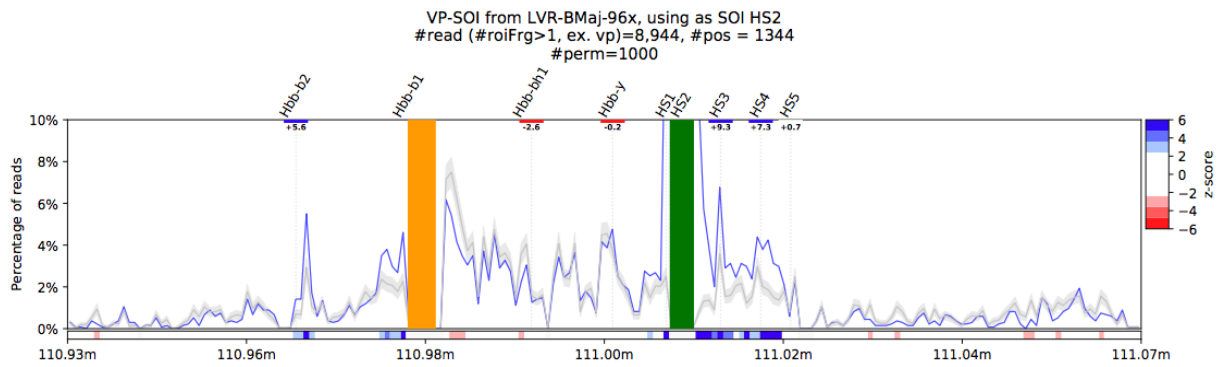


Fig. 10: VP-SOI analysis for the Hbb-b1 gene and HS2 as a site of interest. This figure shows the contact frequency of Hbb-b1 while it is interacting with the enhancer element HS2. This is compared with the background profile generated from the interactions of Hbb-b1 when HS2 is not present in the same read. From this comparison, z-scores are calculated and shown under the profile. **#read(#roiFrg>1, ex. vp)** indicates the number of “informative” reads (concatemers that contain two or more non-viewpoint fragments in the region of interest). **#pos** indicates the number of informative reads that cover the selected SOI. **#perm** indicates the number of permutations used to generate the background profile.

Association test of loci (**atSOISOI**)

The MC-4C pipeline can also perform the multi-way association analysis for every pair of SOIs and visualize the result in a matrix of scores. This analysis can be performed using **atSOISOI** command of **analysis** module. This command produces a heatmap of z-scores where cooperative or mutual exclusive association between each pair of SOIs is represented by positive or negative z-score respectively. An example of such a heatmap is given in **Fig. 11**.

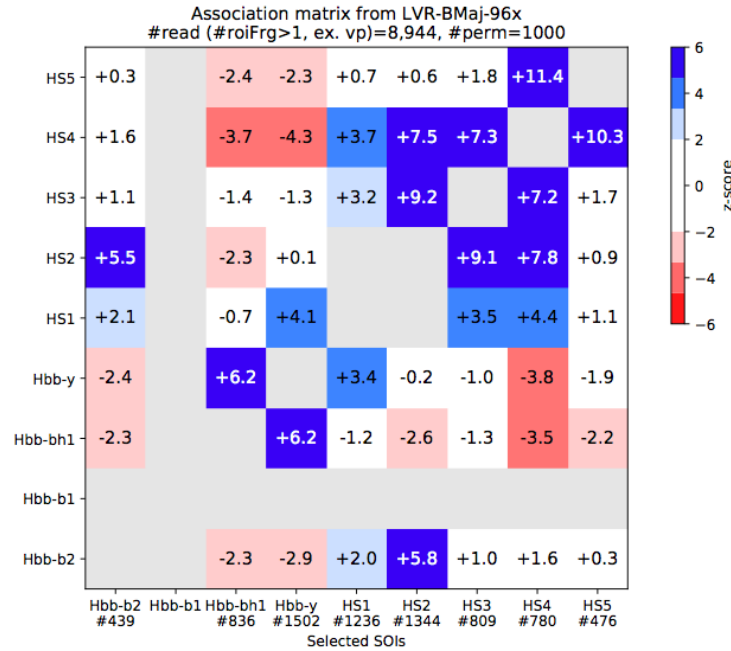


Fig. 11: Association matrix of an MC-4C dataset. The matrix shows the likelihood of a SOI to be found with the viewpoint in the presence of another SOI compared to a case when the given SOI is not present in the read. Each number in a square indicates the corresponding z-score for each pair of SOI. In the bottom axis, the number of reads containing both the viewpoint and a SOI (i.e. number of positive reads) is indicated. **#read(#roiFrg>1, Ex. vp)** indicates the total number of (informative) reads with two or more non-viewpoint fragments in the ROI. **#perm** indicates the number of permutations used to calculate the background profile.

ROI-wide multi-contact analysis (**atAcrossROI**)

The MC-4C pipeline can provide a ROI-wide view of the three-way associations (viewpoint + 2 other area in the ROI). This view is useful when sites of interest in ROI are not well characterized. This analysis can be performed using **atAcrossROI** command of **analysis** module. This command produces a heatmap of z-scores where association between each consecutive 3-bins (considered as a SOI) versus all other bins in the ROI is assessed and the results are represented by a positive (indicating cooperative relationship) or negative (indicating mutual exclusive relationship) z-score. An example of such a heatmap is given in **Fig. 12**.

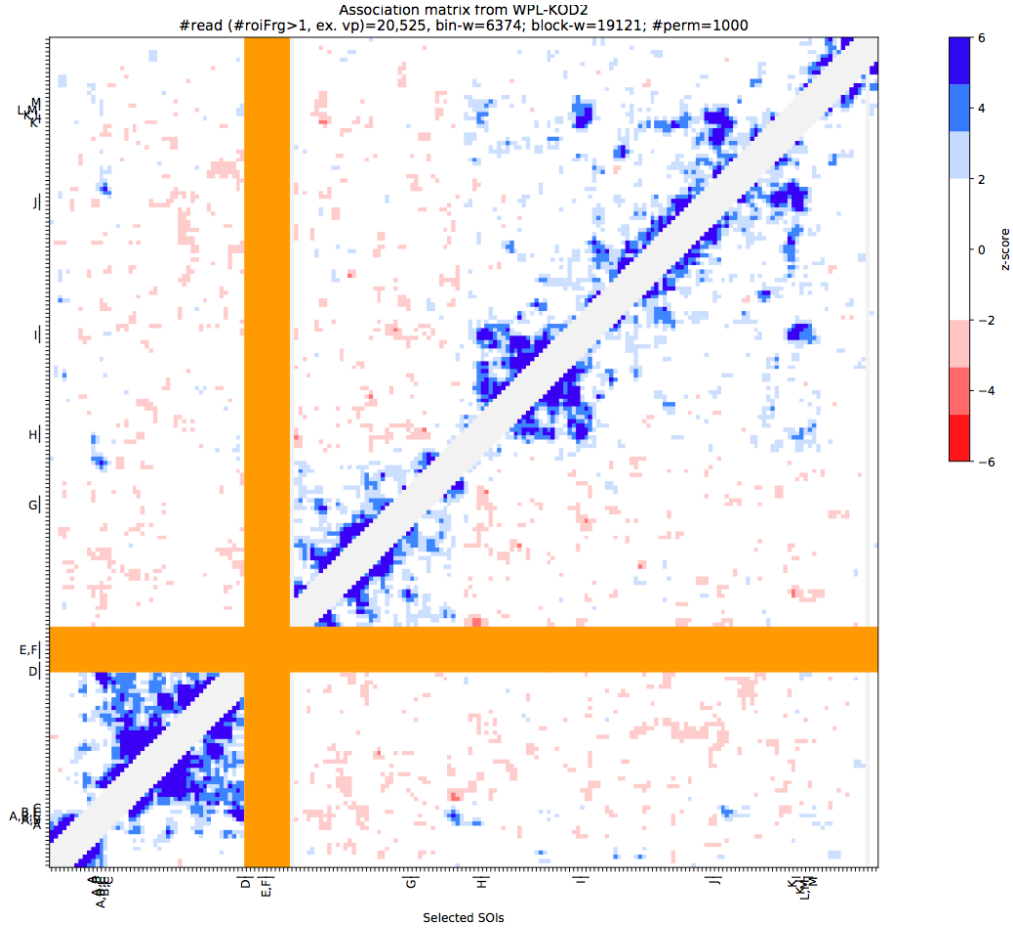


Fig. 12: Association analysis across ROI. Each dot represents association between consecutive 3-bins and a single bin in ROI. Positive z-score (cooperative relationship) is denoted by blue and negative z-score (mutual exclusive relationship) is shown by red color. Some statistics are drawn on the top of the plot; **#read** indicate number of informative reads, **bin-w** indicates the bin width and **block-w** indicates size of SOI (equal to 3 bins). Additionally, number of subsampling is depicted by **#perm**.

Troubleshooting

Pre sequencing troubleshooting

1) *Difficulties obtaining single cell suspensions:* The ability to obtain a single-cell suspension is crucial for homogenous crosslinking, but differs between cell or tissue types. In some cases more stringent treatment is needed, for example by using tissue homogenizers, collagenase treatment or cell straining devices. Spotting a small aliquot of the sample on a microscope slide and viewing the cells under a standard microscope can give an impression of the suspension homogeneity.

2) *Poor digestion efficiency:* Digestion efficiency varies between cell types, and may be inadequate using the standard procedure described here. The first course of action in such cases is to add more restriction enzyme (usually 100U DpnII), and to incubate at 37°C for 3 hours. Repeat the digestion control step, and compare the result with the earlier digestion control (See **Fig. 2c**). If no improvement is visible, the problem is likely due to suboptimal permeabilization. In such cases it may help to recollect the cells by centrifugation and repeating steps 3-10 of the template preparation protocol. Extension of the lysis incubation time (step 3) can also improve digestion efficiency, if incomplete digestions are a recurring problem with a particular cell-type.

3) *Poor ligation efficiency:* If ligation fails to show a clear upwards shift of the template on an agarose gel, the troubleshooting is similar to that of poor digestions: first course of action is to add 40 units of Ligase enzyme, and fresh ATP to a final concentration of 1mM and re-incubating at 16°C overnight. If this does not improve ligation efficiency, repeat step 1-7 of the template preparation protocol to improve cell permeability, and then repeat steps 14 and 15.

4) *Poor gRNA performance:* In some cases gRNAs fail to digest their target DNA, for this reason we recommend first testing them in a small scale on genomic DNA. If gRNAs do not perform well, we recommend designing a different gRNA on the same target region. In cases where gRNAs perform well on gDNA, but not on MC-4C template we have seen improved performance when the MC-4C template was purified using a Qiagen PCR purification kit. However, this does negatively influence the template yield. In this case, split 40 µg's of template over four Eppendorf tubes and purify each aliquot with one column so as not to exceed the maximum loading capacity of the columns.

5) *PCR optimization*: It is essential that MC-4C PCR yields PCR products >1500 bp (see **Fig. 2d**), if this is not the case then further PCR optimization is required. Adjusting primers and annealing temperatures may make a significant difference in large product yield, but in some cases shifting the viewpoint to a neighbouring fragment is necessary. In other cases the MC-4C template may still contain smaller fragments that are favored in the PCR reaction, these can be excluded to some extent by re-doing the 0.5x-0.6x AMPure XP purification. If strong bands are still visible in the PCR smear, as seen in the example in **Fig. 2d**, this indicates that one or multiple gRNAs did not work. If the bands are below the 1.5Kb size-selection threshold, they are still excluded by the pippin size selection, so if the smear does have a significant amount of material >1.5Kb, then it can still be sequenced. However, if enough template is available we recommend troubleshooting the gRNA performance (see troubleshooting #4).

Post sequencing troubleshooting:

1) *Poor size selection*: In the MC-4C protocol a 1.5-8kb interval is used for selecting MC-4C PCR product prior to sequencing. However, after sequencing, we sometimes observed smaller reads. This could be caused by poor size selection procedure. One can assess the efficiency of size selection by inspecting the read size distribution of sequenced library using **QC** module (see **Data quality control** section for details). An efficient size selection should result in a normal distribution of reads sizes in the 1.5-8kb interval. An example of such a distribution is given in **Fig. 13a**. Improper size selection can result in a library that still contains many small reads. An experiment where small reads were unexpectedly sequenced is shown in **Fig. 13b**. Running the MC-4C pipeline on a sample with poor size selection, will result in excessive read filtering, as a high percentage of reads will contain <3 fragments. However, the remainder of reads can still be analysed as usual.

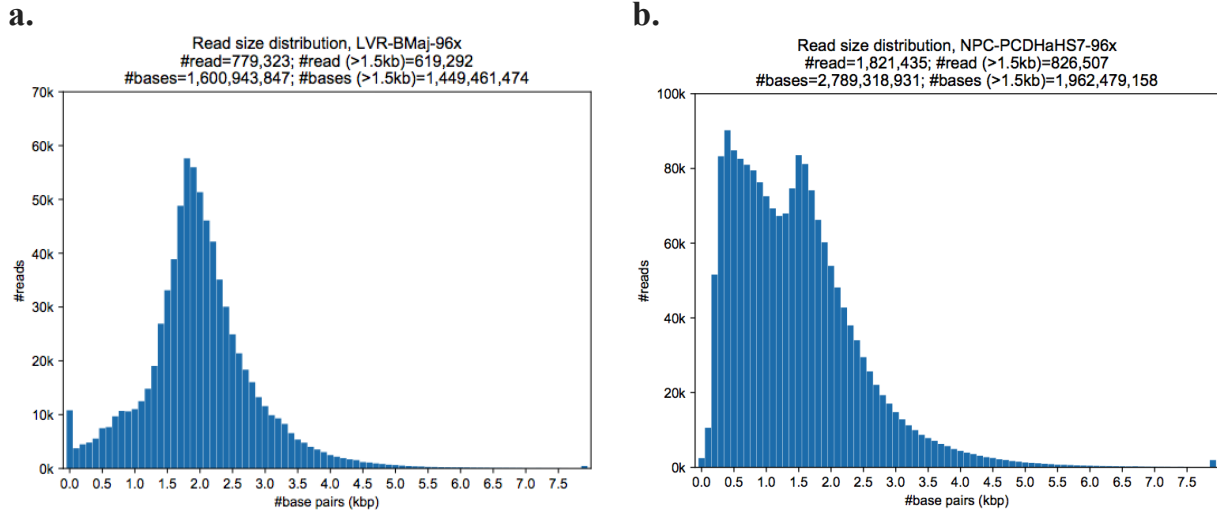


Fig. 13: Read size distribution of an MC-4C experiment. *The size distribution of raw reads should approximate a normal distribution of sizes, where most reads are 1.5-8 kb in length. **a.** Read size distribution of reads with an successful size selection. **b.** Read size distribution of reads with poor size selection. This plot indicates that many small reads are sequenced.*

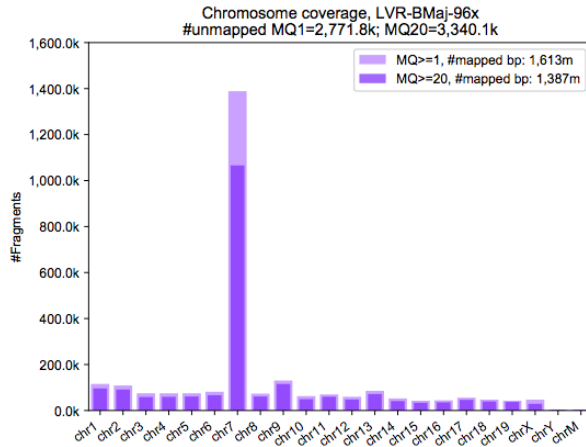
2) Off-target enrichment of MC-4C primers

The specificity of primers can be assessed by inspecting the chromosome coverage plot. Such a plot can be produced by calling “chrCvg” that is implemented in **QC** module:

```
$ python ./mc4c.py QC chrCvg BMaj-test
```

When a high percentage of reads is mapped to non-viewpoint chromosomes, we recommend revising the used primers to either shift them to different positions on the same viewpoint, or to design primers for different a fragment close to the intended viewpoint.

a.



b.

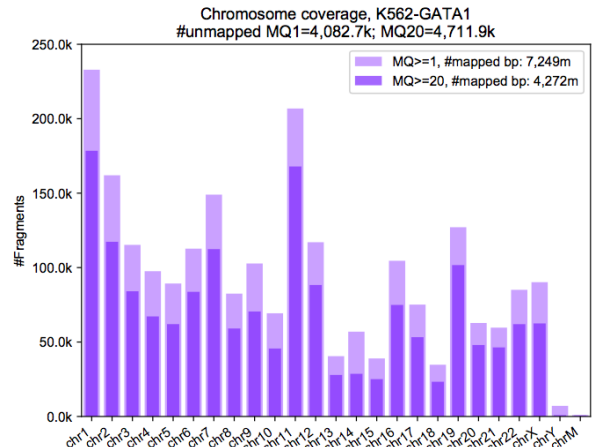


Fig. 14: Chromosome coverage of an MC-4C experiment. *A single chromosome should show a substantial coverage while others are minimally covered. **a.** Chromosome 7, the viewpoint chromosome, is highly covered compared to other chromosomes. **b.** Coverage in viewpoint chromosome (chrX) is not substantially different than other chromosomes, indicative of non-specific primers.*

3) High UMI duplicity

A high UMI duplicity ratio for an experiment (i.e. >20, see **Fig. 15**) indicates a low library complexity. This may be caused by poor PCR performance, which can have several underlying causes. Firstly, the template may not be properly circularized, which can potentially be mitigated by increasing the amount of MC-4C template per reaction. Secondly, PCR performance may be poor due to primer artefacts (self of hetero complementarity, secondary structures). We first recommend designing new primers (see *PCR optimisation* under **pre-sequencing troubleshooting**).

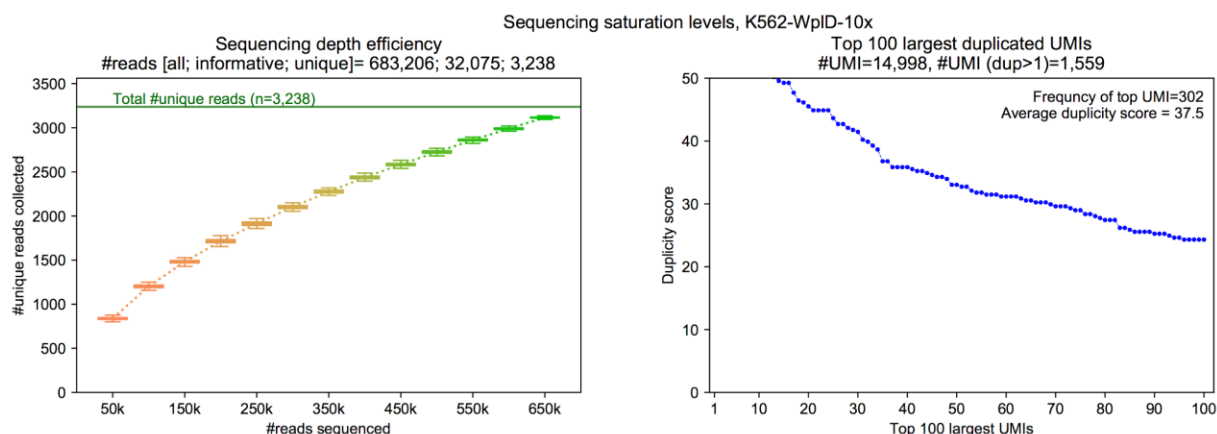


Fig. 15: Sequence saturation plots. *An example of experiment where few UMIs are frequency duplicated (see left panel). In this plot, the average duplicity score (shown on the top right) is calculated to be 37.5 which indicates that diversity of amplified circles in the given experiment is relatively low.*

Author contributions:

C.V. and B.A.M.B. designed and performed experiments, C.V. wrote manuscript and designed figures. A.A. designed and performed the computational analysis, designed figures and wrote the manuscript. P.H.L.K., M.J.A.M.V., C.V.Q. performed ‘C’ methods experiments, R.S. and A.A implemented pipeline in Python, I.J.R. performed and W.P.K. designed and supervised MinION sequencing experiments, G.G. helped with computational analysis, J.d.R. designed and supervised the computational analyses and pipelines and co-wrote the manuscript, W.d.L. conceived and supervised the study and wrote the manuscript.

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Competing Financial Interests

We declare competing financial interests: C.V., B.A.M.B., P.H.L.K., M.J.A.M.V. and G.G. are shareholders of Cergentis. W.d.L. is founder and shareholder of Cergentis. W.P.K. and J.d.R. are co-founder and shareholders of Cyclomics.

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